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A systematic analysis of scan matching techniques for localization in dense orchards

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ABSTRACT

In recent years, different methods have been studied to determine machinery position within a grove, as an alternative for complementing GNSS (global navigation satellite system) information in cases where GNSS signal is occluded. Such a situation can be observed when agricultural machinery travels under dense foliage or on the slopes of mountains. Scan matching techniques arise as a possible solution for localizing the machinery, complementing the absence of the GNSS signal when necessary. However, since key points are difficult to obtain in heterogeneous, unstructured and non-rigid environments (such as orchard plants), the performance of scan matching techniques often decreases in agricultural environments. This paper suggests dividing the point clouds into horizontal and vertical segments to improve the performance of scan-matching methods in orchards. It also examines the best way for registered frames to overlap. We validate the analysis with extensive experimentation in a Fuji apple orchard. The results show that the cumulative localization error in scan matching techniques can be notoriously decreased with selective parts of the orchard, by up to 60%. The experimentation performed herein suggests that the proposed methodology can complement the GNSS navigation in a middle-long path.

1. Introduction

Steer farming vehicles such as tractors and harvesters [26] are used for various activities in the field, such as sensing plant stress, harvesting, yield monitoring or spraying. Although some of those activities require a rough location estimate, others need a centimeter-level accuracy. For example, when spraying is applied to the orchards, a centimeter-level accuracy in the localization is required for the vehicle to spread the pesticide for each orchard based on its status and history [51]. Other tasks, such as harvesting or plant stress monitoring, do not need a centimeter-level accuracy, but they require an accurate and reliable localization system capable of covering large areas.

The global navigation satellite system (GNSS) is commonly used to address the location of autonomous machinery in large agricultural scenarios. The state-of-the-art on GNSS receivers reports a sub-centimeter accuracy in nominal conditions [12]; however, the dense foliage and high vegetation of many orchard environments could make the GNSS antenna readings unreliable due to satellite occlusions [25]. Different sensors, such as dead-reckoning and inertial navigation systems (INS), have been used to assist the navigation in GNSS-occluded areas [61,11,62]. Those sensors are especially useful for short-term navigation; therefore, an extended loss of the GNSS signal would make positioning errors to quickly drift over time. There are some commercially alternatives capable of obtaining an accurate 3D map of the environment in GNSS restricted environments, such as the Leica Pegasus [2], the Viametris bMS3D [16,60], the Robin backpack [35] and the GreenValley LiBackpack [1]. The major disadvantage of these commercial solutions is their high price (several tens of thousands of euros), which makes them too expensive and inaccessible for small-size businesses [59].

The latest advances in sensing technologies, such as LiDAR (light detection and ranging) sensors, have brought attention to agricultural vehicles development, due to their capabilities for phenotyping and or-

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chard characterization [34]. The use of range information to estimate the location of farming vehicles, however, faces several challenges related to the interpretation of the LiDAR-based point cloud information. The terrain roughness and irregularity, for example, can produce disturbances during machinery motion, which can influence the reliability of the range readings [40]. In an orchard environment, one region appears much the same as any other, which causes a lack of visual signatures, and hence, no natural landmarks allow a loop closure [51]. In this context, most of the used techniques to determine the vehicle position within the field are based on using a map built a priori [58,51,50,49]. These types of solutions are often environment-dependent and require expensive hardware, such as D-GPS (differential global positioning system) [58] or UAV [51], to build the map. Other solutions rely on the use of several sources of information.

Several sensors such as odometers, IMU, RGB Cameras, and sonars have been used for completing the navigation of autonomous vehicles in a variety of applications (see [4,5] and the references therein). It is common to see sensor fusion approaches gathering all the information of the aforementioned sensors. To this end, probabilistic approaches (e.g., Extended Kalman Filter, Particle Filter, and Monte Carlo) have been used by the scientific community to perform localization, while others make use of software-based approaches to perform a Simultaneous Localization and Mapping (SLAM) algorithm [53,66]. Two methods arise in these approaches: the probabilistic SLAM and the graph-based SLAM. The EKF SLAM covers the *online* SLAM, which estimates the current robot pose and the map [53]; meanwhile, the graph-based SLAM attempts to solve the *full* SLAM problem [19], but several works also address the online SLAM problem [27,15]. Graph SLAM algorithms are mostly based on scan matching approaches [57]. Scan matching aims to estimate the rigid transformation matrix associated with the vehicle motion. There are at least two ways of estimate robot motion based on scan matching: pairwise scan matching and incremental scan matching. The former matches new arrived scans against the previous scan to find the relative transformation between the poses where the scans have been taken. The latter matches new scans to a maintained global map and is often used in indoor and urban environments, where the map does not change through time. In those man-made environments, there is a significant amount of features such as lines, planes, corners that can be used in the registration approaches, as reported by LOAM [64], LeGO-LOAM [52], and IC3PO [18]. However, in the case of agricultural environments, (1) vehicles are likely to work in an environment filled with trees, grass, which makes the detection of such features unreliable [29], and (2) the map could not be considered static as the foliage/fruits structure will be changing with time. Thus, in this work, we focused on pairwise scan matching, which is the basis of Graph-based SLAM.

Our aim in this work is to use the point cloud information used for farmers in phenotyping activities, and evaluate the usability of that point cloud information for localization purposes. To obtain a complete representation of measured trees when using LiDAR sensors for phenotyping purposes, the LiDAR sensor is usually mounted in vertical orientation [17,42,41,20,21]. However, this sensor orientation reduces the number of key points at longer distances, making the scan matching problem more challenging. In addition, due to the heterogeneity of orchards and the existence of non-rigid elements such as leaves and branches, the amount of valuable key points is further reduced, resulting in a significant decrease in the performance of scan matching techniques even in short path trials. To solve these issues, this paper proposes a new way to deal with these problems by dividing the point cloud into different pieces. This will reduce the search space needed for data association and improve scan matching algorithms.

The remaining of this manuscript proceeds as follows: Section 2 describes the basis of scan matching techniques. In Section 3, we introduce the procedure follows with the problem statement and the notation used in this work. Section 4 presents the methodology. Meanwhile, the experimental results are described and shown in Section 5. The lessons learned are presented in Section 6 and, finally, the conclusions of this work are shown in Section 7.

2. Background

Scan matching techniques aim to align two overlapping scans. To do so, they find a rigid transformation with the translation and rotation between two consecutive scans. Registration algorithms can be, (i) deterministic or (ii) probabilistic. In the earlier nineties, [7] presented the first deterministic procedure for 3D point cloud registration, the well-know Iterative closest point algorithm (ICP). This algorithm minimizes the Euclidean distance between nearest-neighbor points between two scans to find the relative transform. Other deterministic approaches are the Polar Scan Matching (PSM) [23], which changes the distance metric to polar coordinates, and the Iterative Dual Correspondence (IDC) [30], which incorporates a matching range point. Up to date, many techniques have been proposed to improve the ICP performance. For example, in [14], the authors introduced two variants of the ICP based on entropy among points and one method which uses normals to improve the matching stage. An in-depth review of the ICP and some variations in mobile robotics was presented in [43]. Regarding the probabilistic approaches, in [47] it is proposed the Generalized-ICP (GICP), which performed the non-linear minimization considering the maximum likelihood estimator. In a different way of processing data, the Normal Distribution Transform (NDT) [31] describes the use of a voxel-based structure to perform registration. This is, however, one of the main disadvantages, since there is not a validated method of selecting the right voxel size [57]. When the cell is significantly big, the computational time is low, and the accuracy decreases. In contrast, when the voxel size is small, the accuracy increases, but it comes with a high computational cost.

Considering that the correspondences are unknown in real-world scenarios, the aforementioned scan matching variations can only warranty a local minimum. To overcome this problem, an alternative is to expand the region of convergence by initializing the transformation matrix with prior information (e.g., odometry, Inertial measurement unit (IMU)). In realistic operating conditions, however, it is not an easy task to guarantee that an initial estimation is a good one, especially when we are dealing with unstructured outdoor environments [6,34]. Moreover, such information is not always available in farming vehicles. In this work, our aim is to expand the region of convergence by reducing the search space, considering selective parts of the orchard, looking forward to finding the most valuable information for localization purposes. For the purposes of this manuscript, we have selected four vastly known and quietly different scan matching techniques $-ICP_{pt-pt}$(point-to-point), ICP_{pt-pl}(point-to-plane), $GICP$, $NDT-$ as a self-positioning system, as they are the most common scan matcher used for benchmarking [28,48,63,33].

The primary contribution of this work is the evaluation of different segments of the orchard, either horizontal and vertical, for localization purposes. A secondary contribution is the analysis of the overlapping between frames, which is associated with the velocity of the machinery. Our aim in this work was to evaluate the information acquired in the vertical mounting for localization purposes, looking forward to maximizing the use of information while reducing the equipment associated with localization. Thus, the main contribution of this work relies on a deep understanding of the point cloud information of the agricultural environment for localization purposes when using scan matching techniques.

3. Problem statement

For the current manuscript, we consider the world frame as $\{W\}$ and the LiDAR frame as $\{L_k\}$, where k denotes the current scan ($\{L_k\} = \{W\}$ when $k = 1$). Fig. 1 illustrates the position of the sensor in different frames while the platform is moving through the orchards. The

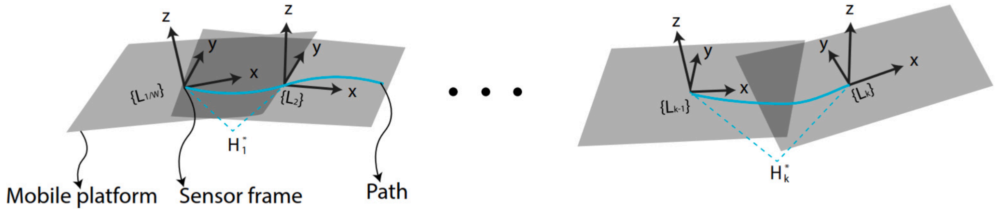

Fig. 1. Movement of the vehicle within the grove, following the path (depicted in cyan). At each LiDAR frame $\{L_k\}$, a point cloud X_k is obtained. The rotation and translation between consecutive frames is obtained with the relative transformation matrix H_k^*.

point cloud in $\{L_k\}$ is X_k and a point, $x \in X_k$ is described by $x_k^L \in \mathbb{R}^3$. At $\{L_k\}$, the data of both sides of the vehicle is extracted as the vehicle is moving within the grove. Considering that the vehicle has a constant velocity, the overlapping between point clouds is analyzed based on the frame selection. For this purpose, the overlapping is introduced as a function of the distance traveled by the vehicle. The twofold selected frames are denoted as: X_k and $X_{k-\ell}$, where ℓ denotes the number of frames ahead. Once X_k is obtained, we assume that such a point cloud can be divided horizontally and vertically in different segments. We refer to the point cloud of horizontal and vertical levels as X_k^{level}. When the vehicle moves, it is desired to find the relative transformation H^*, which aligns X_k with $X_{k-\ell}$. To this aim, a transformation matrix, H_k^{Level} is obtained between the point clouds of the same level for two different frames. The registration procedure takes as input two point clouds: X_k^{level} and $X_{k-\ell}^{level}$, where $k \geq 2$. This procedure is repeated for each level, and as a result the rigid transformation matrix, $H_k^*(\theta)$, with parameters $\theta = [x, y, z, roll, pitch, yaw]^T$ where θ represents a 6-degree-of-freedom rigid transformation, including translation and rotation) is applied to X_k to align it with $X_{k-\ell}$, as $H_k^*(\theta, X_k)$. Such transformation is obtained through an iterative procedure, which consists of a maximum number of iterations, I_{max}, and an error threshold, ϵ. After the relative transformation H_k^* is obtained, the transformation H_k^W from $\{W\}$ to frame $\{L_k\}$ is computed by concatenation of each relative transformation.

3.1. Scan matching

Commonly, when aligning two overlapping scans, the scan that serves as a reference is named as the target, T, and the scan which is translated and rotated is named as source, S. For convenience, let $S \leftarrow X_k$ be the source point cloud and $T \leftarrow X_{k-\ell}$ the target point cloud. Despite the fact that several variations of scan matching techniques have been proposed during the past decade [55], many of those approaches have been tested on specific conditions [28,8,3,67,46,37] and very few in highly unstructured environments [38,44,32,20,39]. For this work, we have selected the four most compared algorithms in unstructured environments; nevertheless, our methodology can be used with any other variation. Below it is shown a brief review of the registration algorithms used in this work.

3.1.1. Iterative closest point and variants

The Iterative Closest Point (ICP) algorithm considers the raw data from both the target and source point clouds. Various metrics can be employed to measure the distance between point sets, such as point-to-point and point-to-plane metrics (refer to Tao et al. [56], Pomerleau et al. [44] and the references therein). The point-to-point method minimizes the sum of squared distances between each pair of corresponding points. On the other hand, the point-to-plane method minimizes the distance between points and the tangential planes at the corresponding nearest points. Correspondences between $H(\theta, S)$ and T are determined using the nearest neighbor criterion $Y_j = \text{ClosestPoint}(H(\theta, S), T)$. The Euclidean distance is used to compute the closest point. Furthermore, the transformation matrix is obtained by minimizing the L_2 error, as shown in Eq. (1).

$$\theta \leftarrow \arg\min_{\theta} \sum_{j=1}^{n} \eta_j \|Y_j - H(S, \theta)\|^2, \tag{1}$$

where η_j represents the surface normal of each point's neighborhood and is utilized in point-to-plane registration. The minimization of Eq. (1) is performed using a least-squares method [7].

In Segal et al. [47], a generalization of the ICP algorithm, known as Generalized-ICP (G-ICP), is presented. G-ICP incorporates a probabilistic framework where covariance matrices are assigned to each point from both point clouds. This approach assumes that $s_i \sim \mathcal{N}(\mu_i^s, \Sigma_i^s)$ and $t_i \sim \mathcal{N}(\mu_i^t, \Sigma_i^t)$ are drawn from independent Gaussian distributions. It is notable that correspondences are also computed using Euclidean distance. For an arbitrary rigid transformation H, $d_i^{(H)} = t_i - H(\theta, s_i)$. If H^* is the correct transformation, then $t_i = H(\theta, s_i)$. Consequently, $d_i^{H^*} \sim \mathcal{N}(0, \Sigma_i^t + H^* \Sigma_i^s (H^*)^T)$. When employing the maximum likelihood estimator to iteratively compute H, the parameters are obtained as shown below:

$$\theta = \arg\max_{\theta} \prod p(d_i^H) = \arg\max_{\theta} \sum_i \log(p(d_i^H)).$$

This can be simplified to:

$$\theta = \arg\min_{\theta} \sum_i d_i^H (\Sigma_i^T + H \Sigma_i^S H^T)^{-1} d_i^H.$$

3.1.2. Normal distribution transform (NDT)

The Normal Distribution Transform (NDT) [31] divides T into voxels and assigns a normal distribution to each voxel with mean u_i and covariance Σ_i. The objective is to find the transformation parameters that maximize the likelihood, p_i, of points from S lying on T. To achieve this, correspondences between points in S and their respective voxels in T are obtained as follows:

$$\tilde{p}(p_i) = -d_1 \exp\left(-\frac{d_2}{2}(p_i - \mu_i)^T \Sigma_i^{-1} (p_i - \mu_i)\right),$$

where $d_3 = -\log(c_2)$, $d_1 = -\log(c_1 + c_2) - d_3$, $d_2 = -2\log\left(\frac{-\log(c_1 \exp(-1/2) + c_2) - d_3}{d_1}\right)$, and c_1 and c_2 are constants related to the cell size. Specifically, c_1 and c_2 can be found by requiring that the probability mass distribution above, equals one within the space spanned by the cell (see [31] for more information). Thus, the NDT score function is defined as $s(\theta) = -\sum \tilde{p}(H(S, \theta))$. The Newton method is used to optimize $s(\theta)$. The Newton optimization method solves the equation $\mathbf{H}\Delta\theta = -\mathbf{g}$, where $\mathbf{H}$ is the Hessian matrix and $\mathbf{g}$ is the gradient vector of the score. It is noteworthy that NDT requires a system with high computational capability, and its performance is directly related to the voxel size.

4. Methodology

Fig. 2 shows a layout of our proposal. Briefly,

- We consider a vertical mounting of the sensor to obtain a more detailed representation of the orchard. This procedure is a common practice when using LiDAR for geometric characterization of vege-

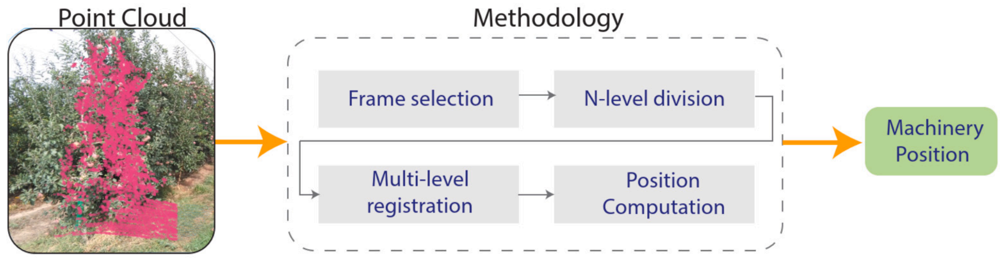

Fig. 2. Pipeline of the methodology. The methodology consists in four main steps; first, the level of overlapping is defined; second, the segmentation is selected (horizontal or vertical); third, a multi-level registration is computed; finally, the estimated position of the machinery is calculated.

tation [17,42,41,20]. The experimental set-up is presented in detail in Section 5.
- The vehicle is moving forward with a constant velocity. The effects associated with skid and slip of the wheels are not analyzed in this work.
- When the overlapping between registered scans is considerable large, commonly, it is more feasible to minimize the registration error. However, if the point cloud is somehow noisy, it can cause the convergence to drop rapidly to a local minimum. Given that agricultural vehicles move at low speeds (0.5 ms^{-1} in this work) and due to the spareness of the point cloud, these effects will appear when associating two consecutive point clouds. For this reason, once a point cloud is acquired, our first step is to manipulate the desired overlapping between registered frames. This process is referred as frame selection and it is described in Section 4.1.
- We propose to split the orchard point cloud into horizontal and vertical segments, achieved through physical height or width division. Since the readings of each segment will be similar between them but dissimilar to other groups, each segment can be considered a salient structural feature of the environment. Thus, a scan matcher could take profit from those structures. Moreover, as the search space for data association is reduced, the scan matching algorithms execution time will also be reduced. Section 4.2 presents how we split the data of each frame in multiple segments.
- Section 4.3 describes how to perform the registration considering multiple segments and to compute the final position estimation.

All algorithms used in this work were developed in Matlab (The MathWorks, Inc. Natick, Massachusetts, U.S.A.) under a Linux operating system. Each of the stages mentioned above is explained in detail as follows.

4.1. Frame selection

Based on the velocity of the mobile vehicle (constant velocity) and the sensors frequency, different amount of overlapping between two consecutive frames can be obtained. Fig. 3 depicts the acquisition of two LiDAR frames at a different time. As can be seen, the overlapping between the two scans is proportional to the distance between $\{L_k\}$ and $\{L_{k-\ell}\}$.

Hence, we can introduce the overlapping as a function of the distance travel by the vehicle, as shown in Eq. (2).

$$Overlapping = \frac{L_{beam} - \ell \cdot \overbrace{\left(V_r \cdot \frac{1}{f_{LiDAR}}\right)}^{distance}}{L_{beam}} \cdot 100\%, \tag{2}$$

where ℓ is the number of frames ahead, V_r is the vehicle velocity, f_{LiDAR} is the LiDAR frequency, and $L_{beam} = 2 \cdot R \cdot tan(\phi/2)$ with $R = 2.4$ meters and $\phi = 30°$. The results for different number of frames ahead are shown in Fig. 4.

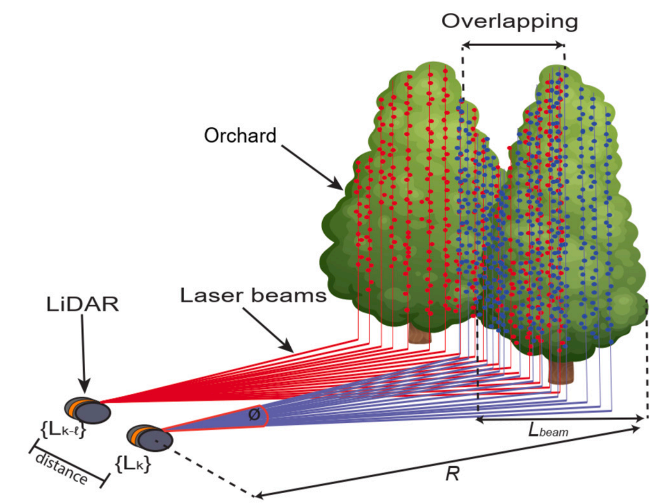

Fig. 3. Data acquisition pipeline using the LiDAR VLP-16. The overlapping among point clouds, X_k and $X_{k-\ell}$ vary according to the distance among LiDAR frames $\{L_k\}$ and $\{L_{k-\ell}\}$; ℓ denotes the number of frames ahead. R and ϕ denote the distance from the sensor to the orchard and the field of view of the sensor, respectively.

As can be seen, the overlapping between the two frames can be manipulated based on the number of frames ahead. During field tests, we found that the estimation begins to diverge with seven frames ahead, which correspond to a 65% of overlapping. Therefore, in this work, we analyzed only the overlapping from one to seven frames ahead for a more meaningful application study.

4.2. N-level division

To focus only on the points, several points of the raw data were removed from the scene as an initial step using the Cloud Compare software. The point cloud information acquired in the orchard is shown in Fig. 5. From this set of points, we removed those points that do not correspond to the orchard and that could lead the scan matcher to diverge. For example, the points that hit the machinery chassis will always be the same, as the LiDAR is mounted on a fixed position. Similarly, antihail net cover and ground points will present a minimum variation in each new scan, leading the scan matcher to estimate a minimum displacement. Since only a few points of the orchards in other rows are captured, such points were also removed using a distance threshold.

4.2.1. Binning

Binning refers to the process of partitioning quantitative attributes into intervals, where the intervals are considered as bins [22]. The process of quantitative binning attributes is common in data mining; common binning strategies are (i) equi-width binning, where the interval size of each bin is the same; (ii) equi-depth binning, where each bin has approximately the same number of tuples assigned to it; and (iii) homogeneity-based binning, where bin size is determined so that

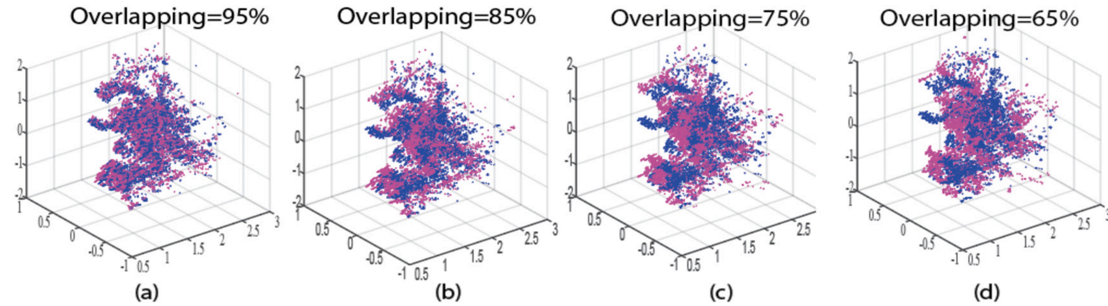

Fig. 4. Overlapping between X_k and $X_{k-\ell}$ for (a) $\ell = 1$, (b) $\ell = 3$, (c) $\ell = 5$ and (d) $\ell = 7$.

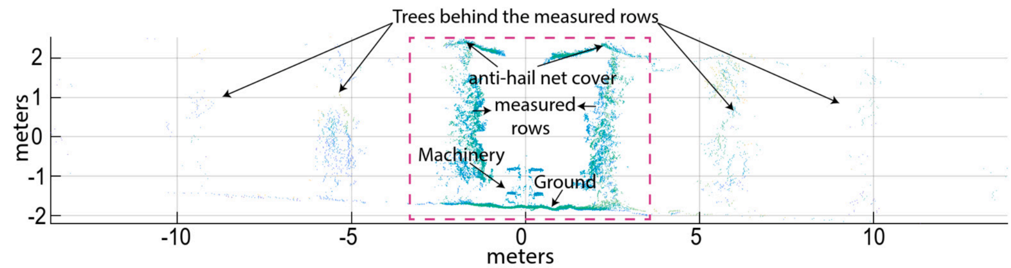

Fig. 5. The point cloud acquisition of a LiDAR, L_k in the Fuji apple environment. The mobile platform circulates along the central alley scanning the two adjacent rows. As can be observed, two additional rows were captured on each side.

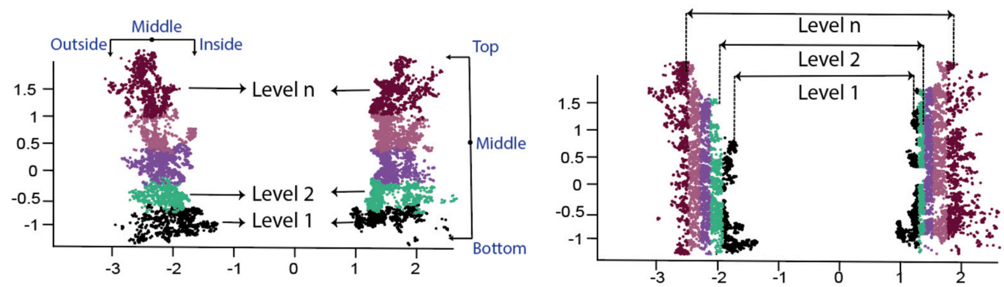

Fig. 6. Horizontal (left) and vertical (right) segmentation for n-levels.

the tuples in each bin are uniformly distributed [22]. In this work, we used equi-depth binning, considering that after this procedure, there is a registration step. Therefore, to accomplish registration with a similar amount of data, we consider different bins of the same amount of points. To obtain the number of bins, $\mathcal{P}$, Sturges formula was used [54], shown in Eq. (3).

$$\mathcal{P} = 1 + \log_2(n), \tag{3}$$

where n is the maximum number of points per scan (∼14000 points after the point removal procedure). Thus, the number of bins, $\mathcal{P}$, was established to 14. As in this work two rows are analyzed simultaneously; each row was divided into seven bins.

4.2.2. Segmentation

Since our goal is to find the portion of the orchard point cloud which contains the most valuable information for localization, we divide the orchard into $\mathcal{P}$ segments; this procedure is described in Algorithm 1. The algorithm is the same for both the horizontal and vertical segmentation, but the point cloud data, X_k, must be sort accordingly to each case.

Algorithm 1 finds the different segments for the orchard point cloud data. In line (1) the algorithm divides the length of the current scan to the number of levels. Then, in lines (2)-(3), the algorithm finds the starting and ending point position for the corresponding level. Finally, in line (4), the selected points are stored in X_k^{level}. Fig. 6 depicts the segmented point clouds after horizontal (a) and vertical (b) divisions were applied. Each segment has the same amount of points; nevertheless, as can be seen, the width of each segment is different. This outcome is mainly related to the density of each zone. Another major point to remark is that the vertical divisions have a smaller spatial distribution than the horizontal divisions. This occurs because the horizontal width of the tree –∼ 1.5 m– is smaller than the tree height –∼ 3 m. For the analysis of the segmented levels, we started with the level one of both sides and increased continuously until level $\mathcal{P}$ was obtained.

Algorithm 1: Segmentation.

input : $[X_k = \{x_i\}_{i=1,\dots,n};\ X_k \in \mathbb{R}^3$
$\mathcal{P}$, level]
output: $[X_k^{level} = \{x_i\}_{i=1,\dots,m/levels};\ X_k^{levels} \in \mathbb{R}^3]$

1 step = size(X_k)/$\mathcal{P}$;
2 $min = (level - 1) * step + 1$;
3 $max = (level) * step - floor(\mathcal{P}/2)$;
4 $X_k^{level} = X_k[min : max, 1 : 3]$

4.3. Multi-level registration and position estimation

Algorithm 2 shows the process for multi-level registration and the pose estimation. First, Algorithm 2 initializes the variables S, T, and C in lines (1)-(3). Second, in lines (5)-(6), the segmentation procedure is applied to the source S, and target T, as described in Algorithm 1. The registration procedure for the corresponding level is applied in line (7). Line (8)-(12) evaluate if the RMSE of the corresponding level has decreased; if so, the optimal transformation matrix is updated. Once all

Algorithm 2: Multi-level registration.

input : $[X_k = \{x_i\}_{i=1,\ldots,m};\ X_k \in \mathbb{R}^3$
$X_{k-\ell} = \{x_j\}_{j=1,\ldots,n};\ X_{k-1} \in \mathbb{R}^3$
$\mathcal{P};\ \boldsymbol{P}_{k-1}]$
output: $[\boldsymbol{P} \in \mathbb{R}^3]$

1 S $\leftarrow X_k$;
2 T $\leftarrow X_{k-\ell}$;
3 C = Inf;
4 **for** $level \leftarrow 1$ **to** $\mathcal{P}$ **do**
// ----- Segmentation -----
5 $S_k^{level} \leftarrow$ Segmentation $(S_k, \mathcal{P}, level)$;
6 $T_k^{level} \leftarrow$ Segmentation $(T_k, \mathcal{P}, level)$;
// ------- Matching -------
7 $H_k^{level} \leftarrow$ Registration$(S_k^{level}, T_k^{level})$;
8 Save $RMSE$ of H_k^{level};
9 **if** $RMSE < C$ **then**
10 C $\leftarrow$ RMSE
11 $H_k^* = H_k^{level}$;
12 **end**
13 **end**
// ------- Compute position -------
14 $\boldsymbol{P}_k^W \leftarrow H_k^* \cdot \boldsymbol{P}_{k-1}^W$

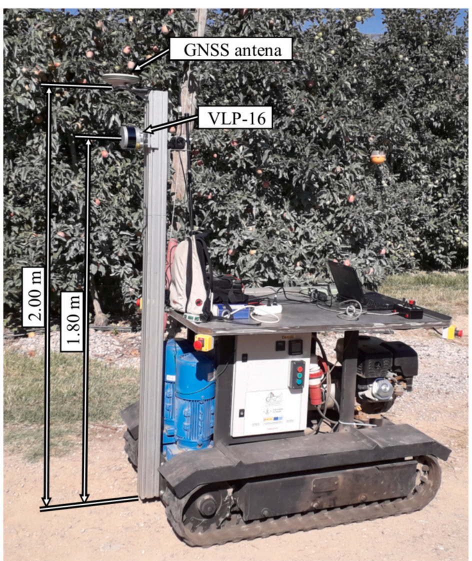

Fig. 7. The mobile terrestrial laser scanner (MTLS). The LiDAR used was a VLP-16 fixed at a height of 1.8 m, and the GNSS antenna was located 2 m above the ground.

levels are evaluated, in line (14) the pose, P_k^W is computed by composing the corresponding transformation matrix H_k^*, as shown in [9].

5. Experimental results

In this work, we use a mobile terrestrial laser scanner (MTLS) equipped with a LiDAR sensor and a real-time kinematics global navigation satellite system (RTK-GNSS), as shown in Fig. 7.

The GNSS rover receiver antenna was installed on top of the mast at a height of 2.0 m. The LiDAR sensor was mounted vertically and positioned at a height of 1.8 m, approximately half the maximum height of the trees studied. This positioning was chosen to ensure consistent detection performance along the tree height. The field test involved moving the MTLS along a straight trajectory parallel to the tree row axis, at a distance of 2.4 m. The forward speed was set to 0.5 ms$^-$1. The LiDAR sensor was a VLP-16 (Velodyne LIDAR Inc., San Jose, CA, U.S.A.), capable of generating a three-dimensional point cloud representation of the scanned scene. The VLP-16 sensor consists of 16 laser beams with a horizontal angular resolution of 2° (30° horizontal FoV). The scanning frequency was set to 10 Hz, corresponding to a vertical angular resolution of 0.2°, yielding a maximum of 28,800 points per scan (acquisition speed of 288,000 points/second). The RTK-GNSS system used was the GPS1200+ (Leica Geosystems AG, Heerbrugg, Switzerland), providing absolute coordinates and UTC time (synchronized with the LiDAR) with a frequency of up to 20 Hz and a precision of approximately 20 mm. All data acquisition was performed using a rugged laptop GETAC V110 (Getac Technology Corporation, Baoshan, Taiwan) with a 64-bit operating system, 8 GB of RAM, and an Intel Core i7-7600U 2.70 GHz processor. LiDAR data was acquired using VeloView 3.5 software (Velodyne LIDAR Inc., San Jose, CA, USA), while GNSS data was collected using a custom-developed LabVIEW (National Instruments, Austin, USA) program that stores the receiver coordinates and UTC time (synchronized with the LiDAR) for each positioning measurement.

The experiments were carried out in a plot of apple trees, Malus domestica Borkh var. Fuji, located at Mollerussa, Catalonia, Spain ($41°36'48.5''N$, $0°51'41.7''E$). Fig. 8 shows the satellite view of the place (a) and a view of the orchards (b). We evaluated one tree row (depicted in green) to find the best combination of overlapping and segmentation levels for each approach. Then, we evaluated the selected parameters in three different rows, defining a long path (approx. 140 meters), a medium path (approx. 70 meters) and a short path (approx. 35 meters).

The experiments were carried out by sub-sampling the target and the source point cloud using an average grid filter following the guidelines presented in [57]. To allow for an efficient convergence of the algorithms, we have fixed the parameters of the algorithms to minimize the error and maximize the performance, following the guidelines described in [45]. The parameters used for registration were defined in: $I_{max} = 100$ and $\epsilon = 0.01$. For the ICP implementation, we used 90% of the paired points with the Euclidean distance. For the NDT, a voxel grid resolutions of 0.5 meter was considered. Regarding the GICP, we used 90 points to construct the covariance matrix.

5.1. Evaluation of the scan matching approaches

To assess the different scan matching techniques, a frame-to-frame registration was performed, considering different levels of segmentation and overlapping. Following previous published work [10,13,14] we have selected the $RMSE$ for describing the error after registration.

The quality of the estimated position was compared with the ground truth data (delivered by the RTK-GNSS), therefore, obtaining the position error, E_{p_k}, as shown below,

$$E_{p_k} = \sqrt{\frac{\left(\hat{P}_k - GNSS_k\right)^2}{2}}. \tag{4}$$

The position error, E_{p_k}, was analyzed for the translation magnitude. We also evaluated the consistency of the heading estimation, following the guidelines presented in [65,24].

In addition, we analyzed the median and the quantiles of the recall-accuracy threshold plots [36,44], which compare the cumulative probability of translation and rotational error against the error magnitude. The registration error ΔT can be defined as follows:

$$\Delta T = \begin{bmatrix} \Delta R & \Delta t \\ 0 & 1 \end{bmatrix} = H H_g^{-1},$$

where H_g denotes the ground-truth transformation matrix, H the corresponding registration solution, Δt the translation vector, and ΔR the rotation matrix. The translation error, e_t, and the rotation error, e_r, can be obtained by,

$$e_t = ||\Delta t|| = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$$

$$e_r = arccos\left(\frac{trace(\Delta R) - 1}{2}\right).$$

By comparing the cumulative probability of translation and rotational error against the error magnitude, we can evaluate the quantiles

Fig. 8. Field testing site. (a) Satellite image with the path followed by the vehicle, and (b) in-situ view of the orchards. The platform was driven following the cyan path.

and the median of the error distribution. Specifically, the quantiles are defined as A50 (i.e., the median), A75, and A95 and correspond to the probability of 0.5, 0.75, and 0.95 of the error distributions, respectively. Based on the quantiles statistics, the precision and accuracy of the registration can be evaluated [44]. The precision is defined as the difference between quantiles and the accuracy is evaluated with the magnitude of the A50 statistic.

First, we are going to analyze the results for the algorithms using the raw data with different overlapping, as explained in Section 4.1. Fig. 9 (a) depicts a qualitative representation of the estimated position; it is shown the estimated path and the true path obtained with the GNSS readings for the evaluation path shown in Fig. 8. As it can be observed, although the tested trajectory was a straight line, all predicted trajectories –using different scan matching techniques– present significant misalignments in scale and rotation. In particular, the ICP in its point-to-point and point-to-plane obtained the worse results in scale, estimating a relative small displacement. The NDT, on the other hand, seems to have more errors on the rotation. The best outcome was obtained with the G-ICP. The results for different levels of overlapping are shown in Fig. 9 (b)-(d). When two frames ahead are considered (b), it is shown a quietly different response for all the algorithms in scale and rotation. Meanwhile, as shown in Fig. 9 (c)-(e) for two, three, and four frames ahead, the performance of all algorithms improves, some in scale, and some in rotation. However, for six and seven frames ahead (Fig. 9 (f)-(g)), only the NDT improves the results of Fig. 9 (a).

Fig. 10 summarizes the results for different number of frames ahead by visual inspecting the results presented in Fig. 9. As can be seen, the outcome can be improved with less amount of overlapping. This happens due to fact that when traversing with low velocities (0.5 ms^{-1} in this work) the distance among points and the spareness of them make the registration to quickly reach a local minimum, therefore it sub-estimates the results. All the approaches analyzed herein can only warranty a local minimum; however, different frames ahead can alleviate the local minimum obtained in a frame-to-frame registration. It is worth to mention that a global minimum can not be guaranteed.

For evaluating the impact on the proposed methodology, first, consider the iterative closest point-to-point version and the horizontal segmentation of two consecutive scans. Fig. 11 shows the RMSE variation after registration (ICP) when one, two, three, four, five, six, and seven segments are used to cluster the data. Notoriously, it can be seen that the original $RMSE$ (one level) continuously decreases until five levels are used. When the sixth and seventh levels are used, the $RMSE$ increase. Similarly to the overlapping results, this outcome suggests that the local minimum obtained in a frame-to-frame registration can be alleviated using only certain orchard information. In order to validate those hypotheses, several experiments were carried out.

5.2. Horizontal division

In order to find which portion of the orchard contains the most useful information for registration, we divided the orchard point cloud into different segments. First, we analyze the horizontal segmentation. Seven different levels of segmentation were tested, starting with two levels and continuously increasing until reaching seven levels. Similarly, the frames ahead (overlapping) increased from one to seven. The results presented herein depicted the mean square error, E_{p_k}, of the total estimated position (the magnitude of the X and Y-coordinate error).

Fig. 12 depicts the results for the four algorithms used in this work considering different overlapping levels (frames ahead) and a different number of horizontal segmented levels. As can be seen, the number of frames varies according to the number of frames ahead. For the ICP point-to-point, it can be observed that the highest cumulative error was obtained with one, six and seven frames ahead; meanwhile, the lowest error was obtained with two and three frames ahead considering six and seven levels of segmentation for both cases. The ICP point-to-plane registration approach seems to perform similarly to the point-to-point case because the highest and the lowest errors were obtained using the same number of frames ahead and horizontal levels. On the other hand, the GICP shows the highest error with six and seven frames ahead. It can be seen that for two, three, four and five frames ahead, the obtained error is smaller than with one frame ahead. From those three cases, the best outcome was obtained with three frames ahead using two and three segmentation levels. For the NDT approach, the results show that an increase in the number of segmentation levels does not improve the performance of the algorithm, conversely to the other approaches. In fact, it is shown that the highest error was obtained with seven frames ahead and seven levels of segmentation. Nevertheless, the error is notoriously decrease using the original data and five frames ahead. A possible reason for this discrepancy is the cell size; even though the size of the cell was relatively small (0.5 m), when analyzing the orchard with more segmentation levels, the number of points in each cell decreases, leading to alignment errors.

Now, let's analyze the information contained in each segmentation level. This means that we are looking at the number of times each segmented level was selected. Fig. 13 shows the levels which contain the minimum RMSE. For example, for one level of segmentation, there is only one column due to the raw data is used. The number of columns is equal to the number of segmentation levels. The figures of each approach were selected based on the results previously discussed; three frames ahead for the ICP_{pt-pt}, ICP_{pt-pl} and GICP, and five frames ahead for the NDT. For the ICP point-to-point (a), it is shown that for six segmentation levels, the results show that the most useful information is contained in the fifth and fourth levels, which correspond to the orchard's middle-top. For the point-to-plane (b), it is shown that for six

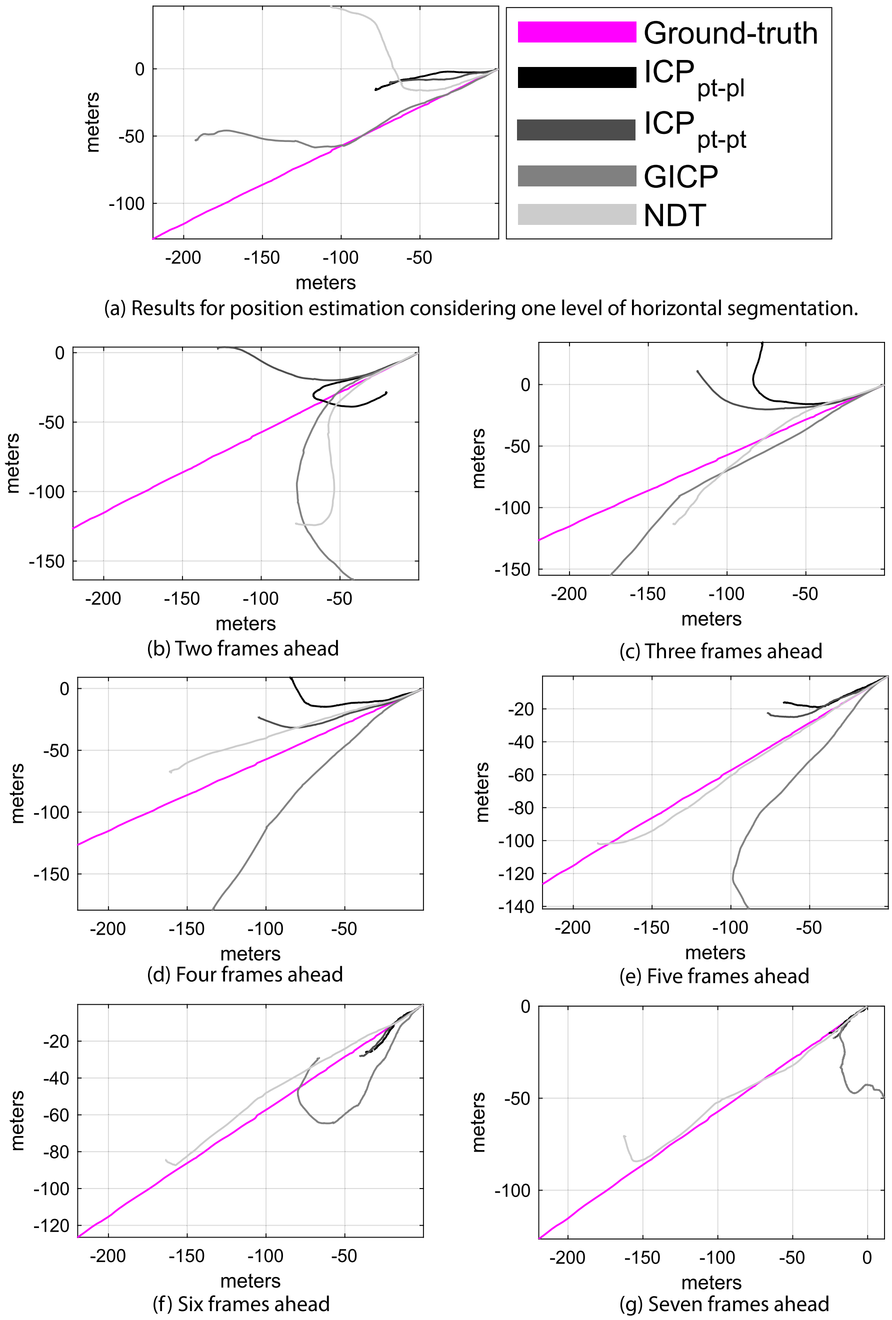

Fig. 9. Results for position estimation based on a frame-to-frame registration, considering different overlapping between consecutive frames.

levels of segmentation, it can be seen that the bottom of the orchard is selected a few times. In contrast, the fifth level is the one with the majority of frames selected, a similar result to the one obtained with the point-to-point version. Regarding the GICP, it is shown that the bottom of the orchard is selected a few times and the middle part of the orchard seems to be selected most of the time. If we analyze the case of three levels, the middle part of the orchard is mostly to be selected, followed by the top of the orchard and only few times the bottom contain the minimum $RMSE$. Finally, for the NDT, it is shown that the middle and the top of the orchard are selected most of the time. However, from the analysis of E_p in all the axis, it was seen that increasing the numbers of segmentation levels, most of the time, increase the error.

5.2.1. *Testing the position estimation within the field*

After analyzing various levels of overlap and segmentation, we evaluate the best results in three different rows of the field (see Fig. 8), corresponding to a long, medium, and short path. Fig. 14 presents the results for each method when performing horizontal segmentation at different overlap levels. Specifically: six segmentation levels and a three-frame look-ahead were used for the ICP_{pt-pt} method; six segmentation levels and a three-frame look-ahead were applied to the ICP_{pt-pl} method; three segmentation levels and a three-frame look-ahead were evaluated for the $GICP$ method; one segmentation level with a five-frame look-ahead was used for the NDT method.

Fig. 14 demonstrates that all four scan matching techniques tend to align closely with the ground-truth data. When comparing these results with the original implementation shown in Fig. 9, it is evident that, across all three paths (in different rows), both scale and rotation of the algorithms have significantly improved. Despite the marked reduction in rotation error, a small drift remains in all approaches, which accumulates along the paths. This drift is particularly noticeable in the long

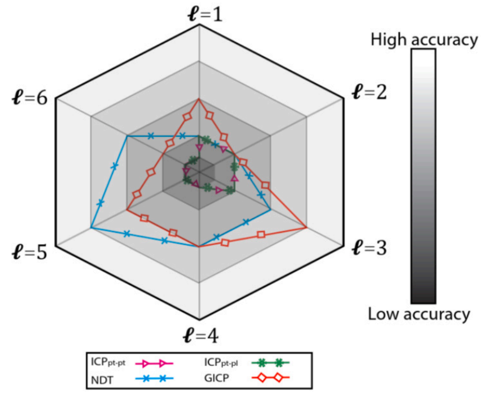

Fig. 10. Qualitative summary of registration approaches with different overlapping levels. The number of frames ahead is denoted as ℓ. High accuracy is obtained when the estimated path is closer to the ground-truth path; meanwhile, low accuracy is obtained when the estimated path is farther to the ground-truth.

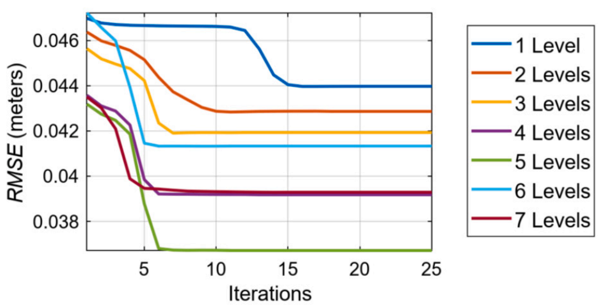

Fig. 11. Cost variation for the iterative closest point in its point-to-point version using a different number of horizontal divisions and two consecutive scans.

path for the NDT method and the medium path for the ICP_{pt-pt} method, where the trajectory deviates from a straight line.

The consistency analysis for the heading estimations is also presented in Fig. 14. The approaches are deemed consistent when the error is bounded by twice their standard deviation, as per [65,24]. While the estimations are consistent for most of the analyzed frames, there are instances where inconsistency arises. At these points, the heading error introduces significant drift in the overall estimation. The key limitation in applying scan matching to estimate the vehicle's pose is its dependence on the previous state. For instance, in the short path, the ICP_{pt-pt} remains consistent throughout (Fig. 14 (f)), resulting in the most accurate estimation (Fig. 14 (e)). In contrast, inconsistencies in the other approaches lead to deviations in their position estimates.

5.3. Vertical division

For the vertical case, as for the horizontal case, seven different levels of segmentation were tested. The overlapping was also analyzed by increasing the frames ahead from one frame to seven. Similarly to the horizontal case, in this section, it is presented the evaluation and testing of the position estimates.

Fig. 15 depicts the results of a different number of vertical segmentation levels. For the ICP_{pt-pt}, it can be observed that there is not a significant difference when the numbers of frames ahead vary. The lowest error was obtained using four frames ahead and using the original data (one level). Similarly, the ICP_{pt-pl} obtained a similar behavior for different number of frames ahead, but the lowest error was obtained with seven levels and two frames ahead. The $GICP$ obtains higher errors with five, six and seven frames ahead and obtains the smaller error considering three frames ahead and three segmentation levels. For the NDT approach, the results show that an increase in the number of segmentation levels does not improve the performance of the algorithm, as in the horizontal case. The smallest error was obtained with the original data (1 level) and five frames ahead.

Fig. 16 shows the number of selected levels which contain the minimum RMSE. The figures of each approach were selected based on the results previously discussed; four frames ahead for the ICP_{pt-pt}, two frames ahead for the ICP_{pt-pl}, three frames ahead for the $GICP$ and five frames ahead for the NDT. For the ICP_{pt-pt} (a), it is shown that for two number of divisions, most of the useful information is on the inside of the orchard; meanwhile, for higher levels of segmentation (number of divisions) the most useful information is located in the outside of the orchard. However, as previously discussed, the lowest error was obtained considering the original data (1 level). For the ICP_{pt-pl} (b), it is shown that for seven levels of segmentation, the majority of frames selected are in the middle outside of the orchard. In the $GICP$ case, for three levels of segmentation, the inside, the middle and the outside of the orchard were selected several times, being the outside the one which was selected most of the times. For the NDT, it is shown that the outside of the orchard is selected most of the time, except when two segmentation levels were considered. However, as in the horizontal case, it was seen that increasing the numbers of segmentation levels, most of the time increases the error.

5.3.1. Testing the position estimation within the field

Similar to the horizontal case, we evaluate the best outcomes in three different rows (long, medium and short paths) of the field. Fig. 17 depicts the outcomes for each algorithm when performing vertical segmentation with different levels of overlapping. Specifically, one level of segmentation and four frames ahead were considered for the ICP_{pt-pt}; seven levels of segmentation and two frames ahead were considered for the ICP_{pt-pl}; three levels of segmentation and three frames ahead were considered for the $GICP$; one levels of segmentation and five frames ahead were considered for the NDT.

The results of Fig. 17 indicates a higher cumulative error than in the horizontal case. Only the $GICP$ in the long path does not obtain a rotational error; meanwhile, in the other cases, all the algorithms depict a significant error in rotation along with some inconsistent estimations. For example, in the long path, the ICP_{pt-pt}, ICP_{pt-pl}, and the NDT trend to estimate a slight curvature instead of a straight line. It can also be observed that the ICP_{pt-pt} inconsistent estimation between the frames 2500 and 2800 produces a complete turn in the position estimation nearly to the end. When comparing the results to those obtained with the horizontal segmentation, it can be seen that the vertical resolution obtains less accurate results. One of the main reasons is the spatial distribution of the points, which for the vertical case is smaller than for the horizontal case, as explained in Section 4.2.2. For such reason, most of the algorithms obtain the lowest error using few segmentation levels.

5.4. Recall-accuracy threshold plots

Fig. 18 and Fig. 19 show the cumulative probability of translation and rotational errors considering all the individual frames in the long, medium and short path. Such figures present the proportion of outcomes that lie beneath a given error. It is worth mentioning that for each algorithm, the vertical and horizontal combination denoted in the legend is different as explained in 5.2.1 and 5.3.1. For the translation error in Fig. 18, it is evident that the vertical and horizontal combinations reduce the cumulative error for all the algorithms, as those curves reach the maximum before the original implementation. In particular, it can be seen that the horizontal combination in all the cases obtains a better outcome, reaching a maximum of around 0.1 meters of a translation error.

For the rotational error, it can be seen a very similar response for the horizontal and vertical combinations. Both outperformed the original implementation of the four registration approaches. The horizontal

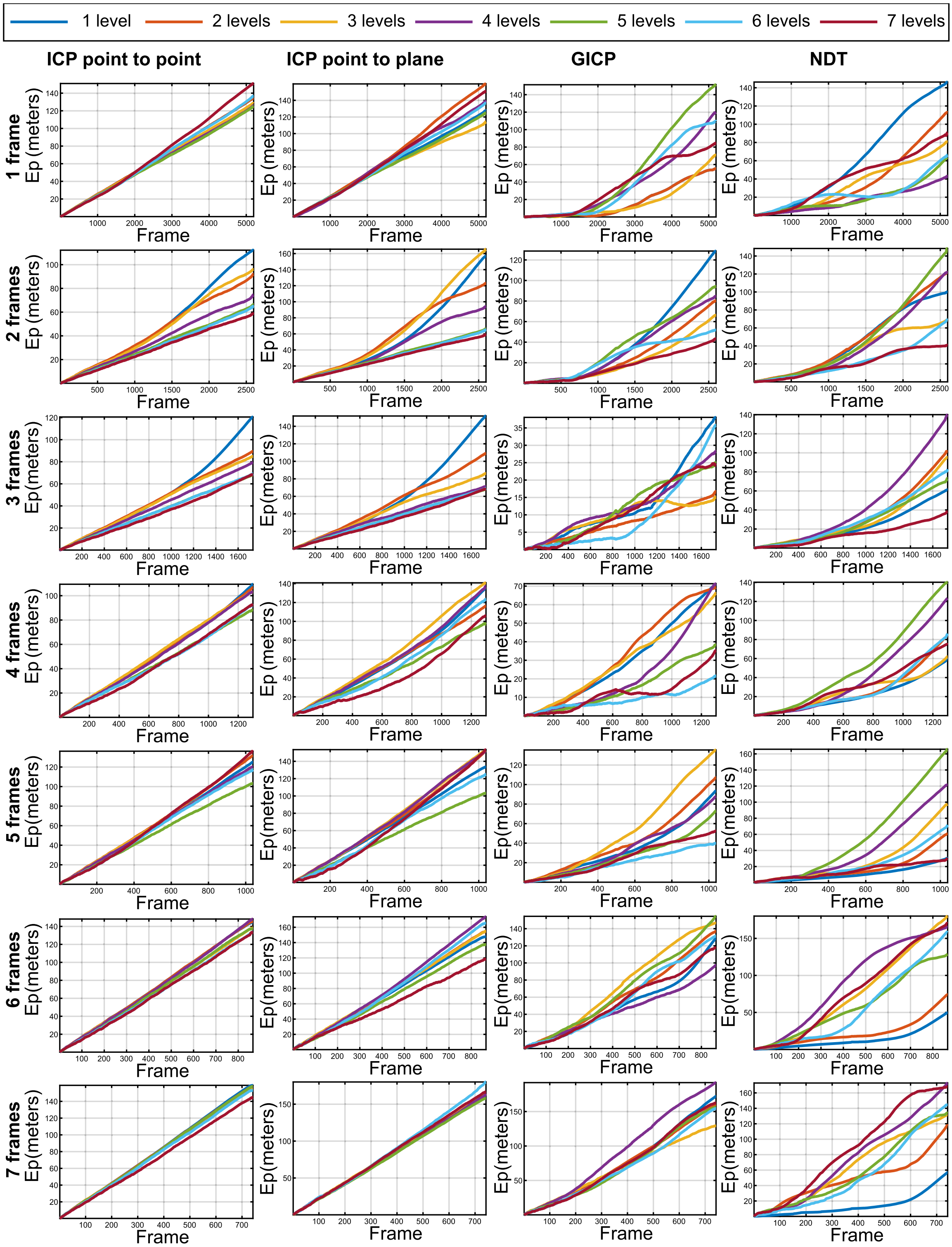

Fig. 12. Summary of the error in the absolute position estimation considering different overlapping and horizontal segmented levels. Each color denoted the number of levels and each row the number of frames ahead.

and vertical combinations reach a maximum close to 0.03 radians of rotational error in the four cases. The variation in the heading of the vehicle between consecutive frames is small, as the vehicle traverses the orchards almost in a straight line.

Table 1 summarizes the results for the quantiles in translation and rotation error for the four registration approaches. For the translation case, it can be seen that the NDT is the most accurate algorithm for horizontal and vertical segmentation since it has the lowest A50 value (2.14×10^{-2} meters). Considering the difference between the A95 and A50, it can be seen that ICP point-to-plane and the NDT obtain a difference of 2.59×10^{-2} meters and 2.54×10^{-2} meters, respectively, making them the most precise algorithms. Meanwhile, the lowest precision in translation was obtained with the GICP approach. When analyzing the rotation error, it can be seen as an improvement with respect to the original implementation. The most accurate algorithm in the rotation was the NDT, which has a value of 0.35×10^{-2} meters of error for the

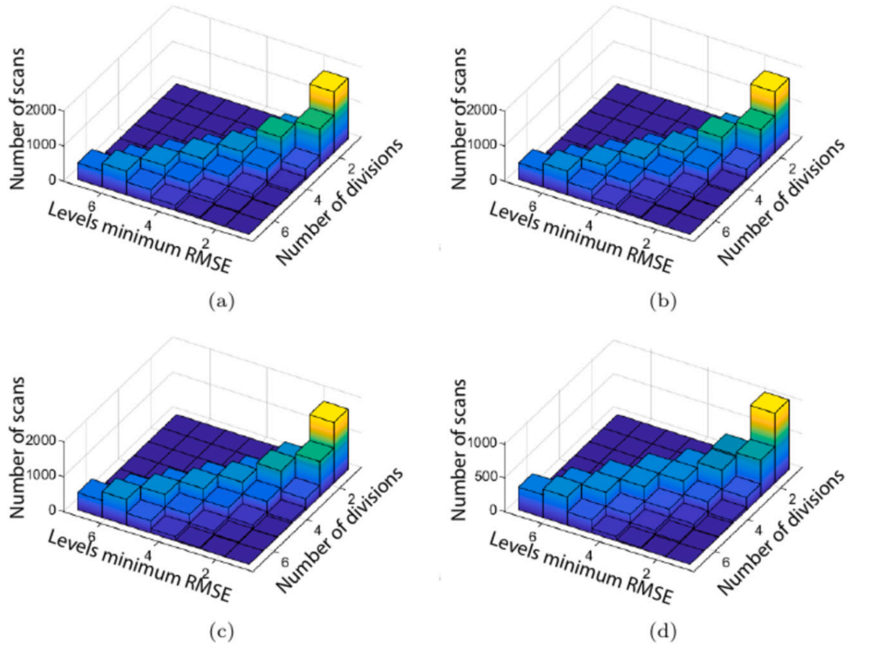

Fig. 13. Analysis of the levels with minimum $RMSE$ for (a) ICP point-to-point, (b) ICP point-to-plane, (c) GICP and (d) NDT. The figures on the left depict the number of frames selected for each level of horizontal segmentation and the figures on the right a quantitative representation of the mean and variance of the $RMSE$.

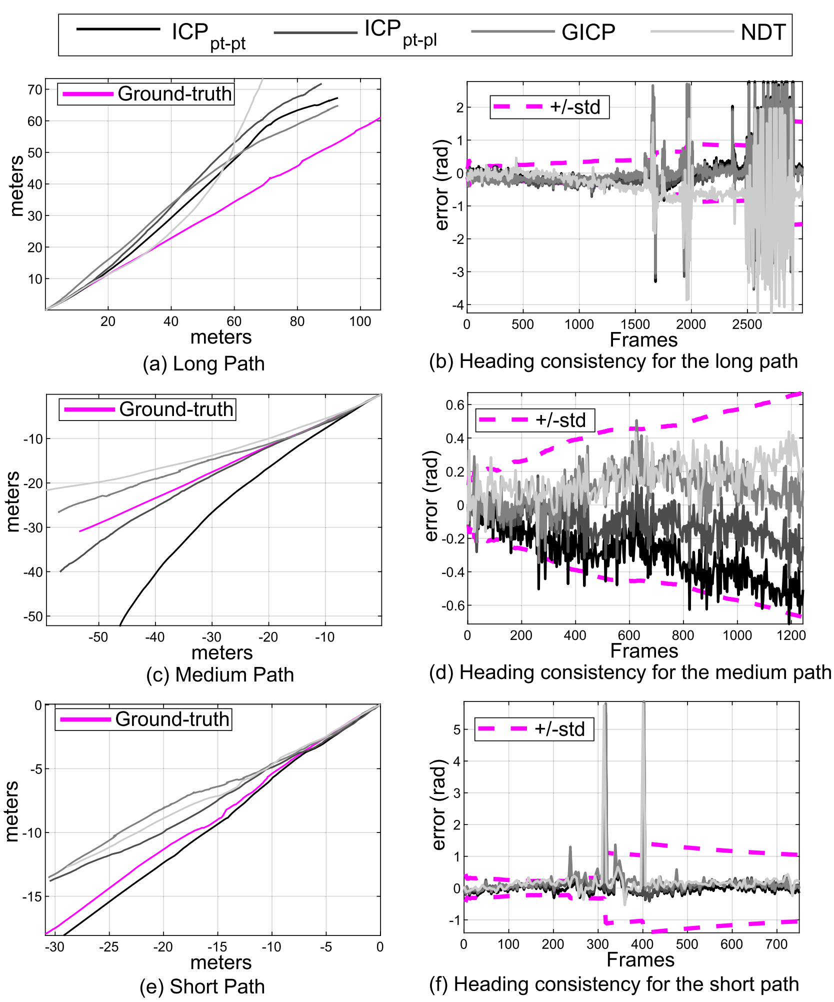

Fig. 14. Evaluation of the proposal in different zones of the field, considering a long path (top), medium path (middle) and short path (bottom). On the left the results for position estimation and on the right the results of the consistency analysis.

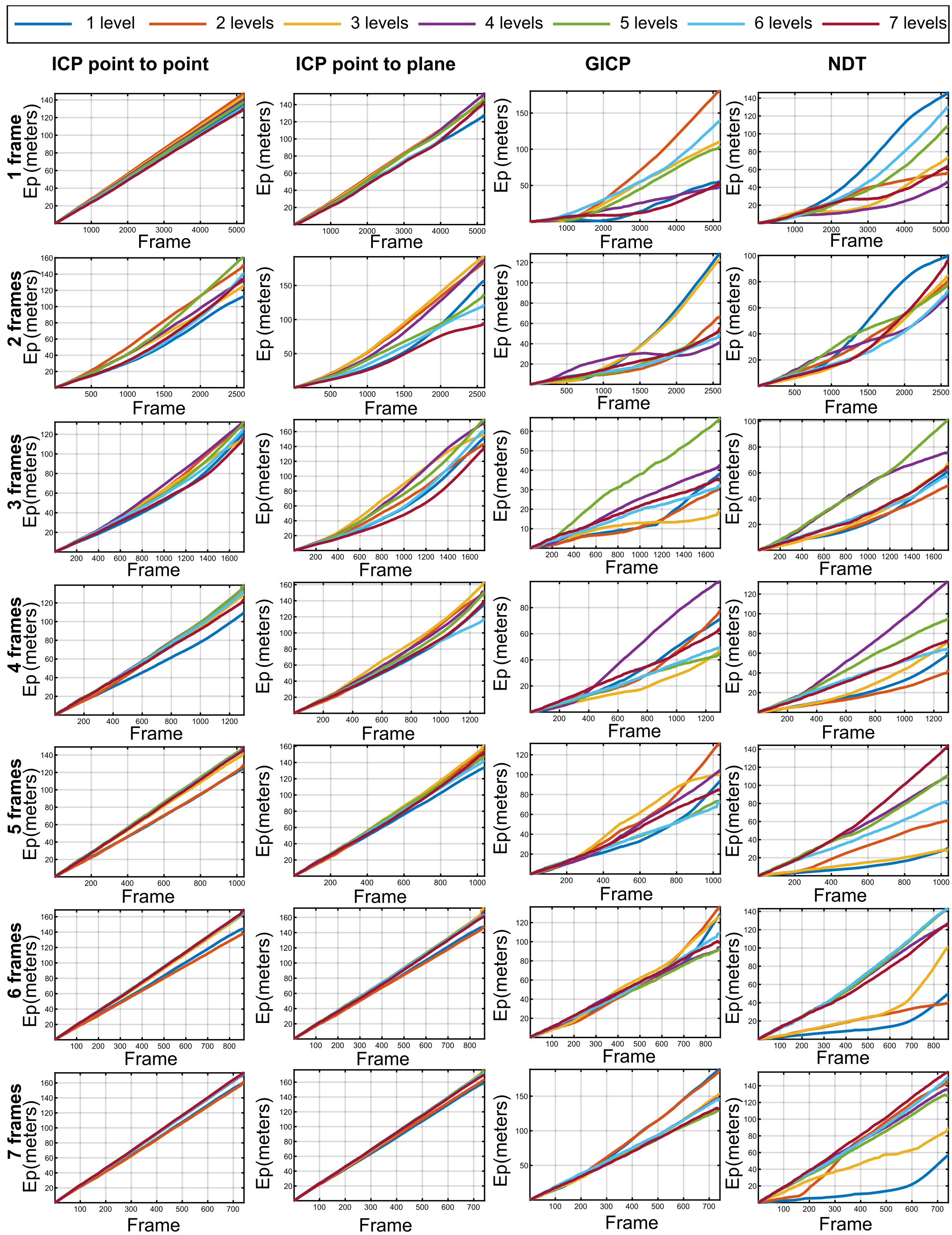

Fig. 15. Summary of the error in the absolute position estimation considering different overlapping and vertical segmented levels. Each color denoted the number of levels and each row the number of frames ahead.

A50 statistic. Finally, the most precise algorithm in the rotation was the ICP point-to-point since it obtains the smallest difference (0.9×10^{-2} radians) between the A95 and A50 statistics. Although the results are very encouraging, the nature behind such behavior still needs of further study. Up to date, such scan matching approaches have only been implemented in structured environments (as stated in Section 3, and the agricultural scenario presents point cloud that is challenging due it its randomness spatial distribution.

6. Lessons learned

During the development and experimental validation of the proposed methodology, several lessons were learned.

- Estimating position using scan registration in unstructured environments remains an open field of study. In agricultural scenarios, there are many challenges to providing a general solution for scan

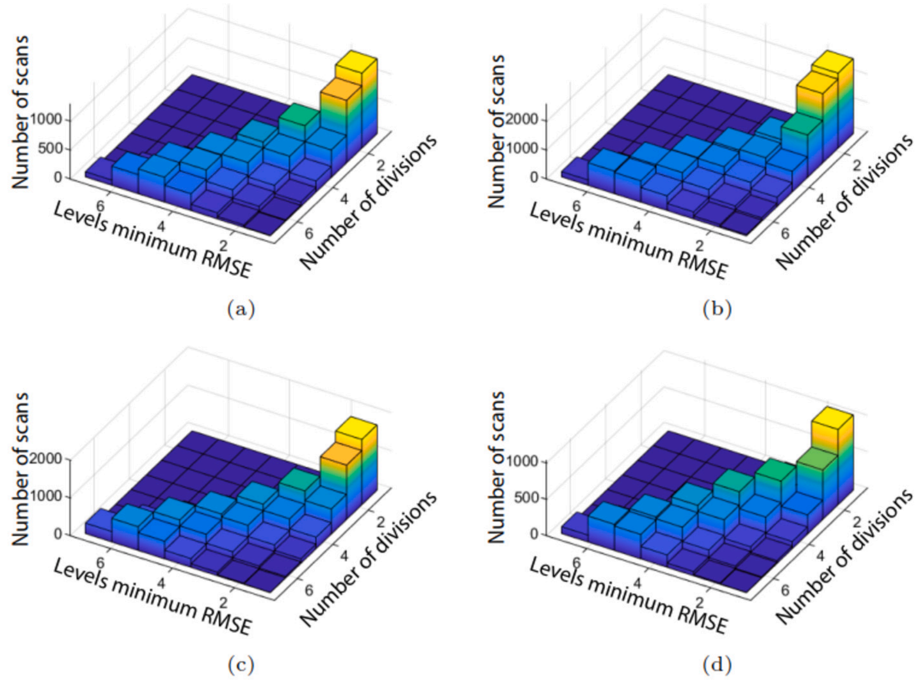

Fig. 16. Analysis of the levels with minimum $RMSE$ for (a) ICP point-to-point, (b) ICP point-to-plane, (c) GICP and (d) NDT. The figures on the left depict the number of frames selected for each level of vertical segmentation and the figures on the right a quantitative representation of the mean and variance of the $RMSE$.

Table 1
Statistics of the recall-accuracy threshold plots. A50, A75 and A95 denote the probability of 0.5, 0.75 and 0.95 of the errors distribution.

Wavelet	Registration	Translation (meters)			Rotation (radians)		
		A50	A75	A95	A50	A75	A95
Horizontal	ICP_{pt-pt}	4.37e-2	5.40e-2	7.08e-2	0.41e-2	0.68e-2	1.31e-2
	ICP_{pt-pl}	2.33e-2	3.26e-2	4.92e-2	0.43e-2	0.77e-2	1.51e-2
	$GICP$	8.92e-2	9.97e-2	11.6e-2	0.39e-2	0.75e-2	1.53e-2
	NDT	2.14e-2	3.11e-2	4.68e-2	0.35e-2	0.64e-2	1.34e-2
Vertical	ICP_{pt-pt}	4.59e-2	6.97e-2	11.51e-2	0.42e-2	0.75e-2	1.67e-2
	ICP_{pt-pl}	4.59e-2	6.97e-2	11.5e-2	0.42e-2	0.75e-2	1.67e-2
	$GICP$	9.49e-2	11.71e-2	15.78e-2	0.55e-2	0.99e-2	2.05e-2
	NDT	2.14e-2	3.11e-2	4.68e-2	0.35e-2	0.64e-2	1.34e-2
Original	ICP_{pt-pt}	8.29e-2	11.6e-2	17.1e-2	0.88e-2	1.48e-2	2.63e-2
	ICP_{pt-pl}	8.28e-2	11.7e-2	17.1e-2	0.89e-2	1.54e-2	2.74e-2
	$GICP$	11.9e-2	16.9e-2	24.7e-2	5.47e-2	9.42e-2	16.3e-2
	NDT	12.1e-2	16.8e-2	24.9e-2	5.63e-2	9.62e-2	15.9e-2

matchers, such as handling massive amounts of data, environmental variability, the reliability of range readings, the sparse structure of orchards, and varying climate conditions. In this work, we used four different rows of the environment, one for evaluation and three for testing, aiming to provide a comprehensive evaluation of scan matchers.

- Point association is a critical factor in scan registration. It can be improved using the multi-level registration proposed in this work; however, evaluating different levels is necessary to find the optimal combination that enhances the scan matcher's performance.
- The velocity of the mobile platform directly affects the overlap between consecutive scans. As travel velocity increases, overlap decreases. In this study, we analyzed overlap considering various frames ahead since the vehicle moved at a constant speed. The results show that considering multiple frames ahead can help mitigate local minima when the overlap is at least 65%.
- When evaluating overlap and segmentation simultaneously, registration was performed using the segment that achieved the minimum $RMSE$. For the horizontal case, the results suggest that the most valuable information is located in the middle-top of the orchard. In contrast, for the vertical case, the most useful information was found outside the orchard when five, six, and seven divisions were considered, and inside the trees when two, three, or four segmentation levels were used.
- Positioning estimates for all algorithms, in both the horizontal and vertical cases, show improvements in scale and rotation even for long paths. However, small rotational errors may accumulate over time, causing the estimates to drift from the ground truth. This effect was more pronounced with vertical segmentation. In both cases, the estimation became inconsistent at certain points, resulting in significant drift from the path. This is a major limitation of relying on a single information source, such as LiDAR, because the vehicle does not revisit the same location and cannot recover from registration errors.
- We did not analyze the vehicle's turning between rows in this work because, during a turn at the end of a row, the vehicle begins to lose half of the information (from one side of the orchard). At some

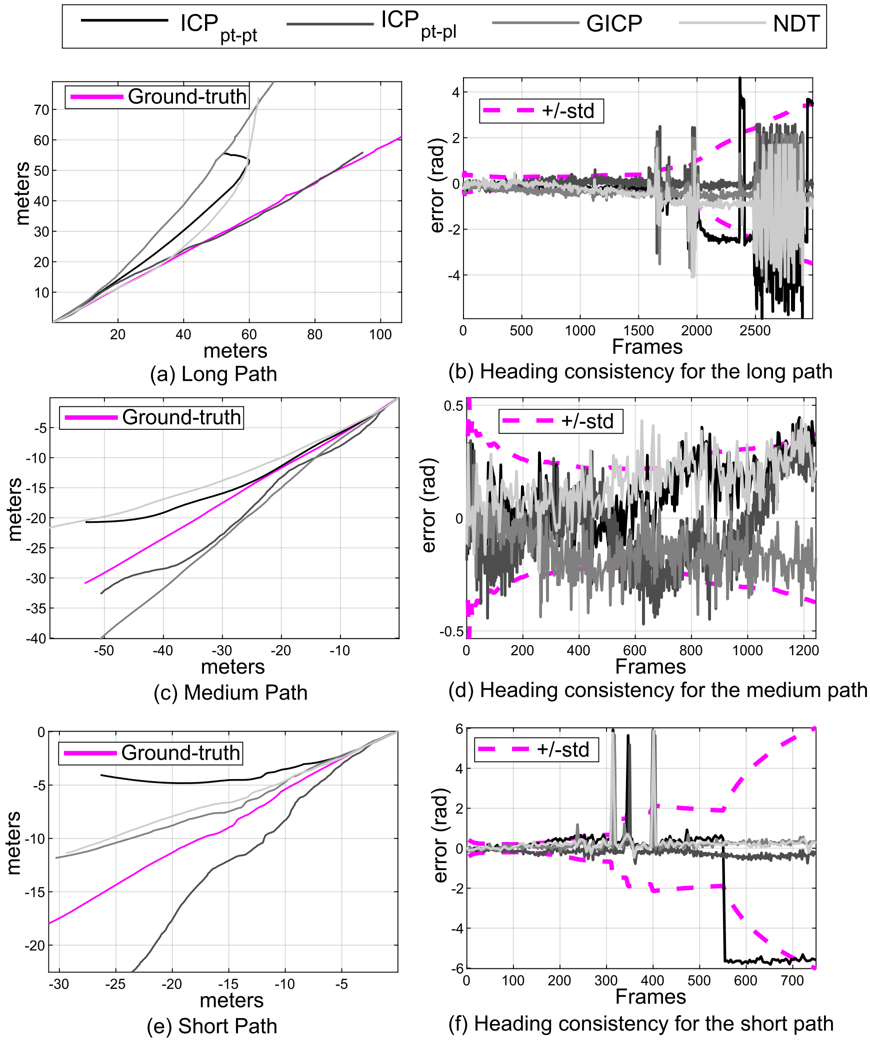

Fig. 17. Evaluation of the proposal in different zones of the field, considering a long path (top), medium path (middle) and short path (bottom). On the left the results for position estimation and on the right the results of the consistency analysis.

point, the Multi-Level Tree Segmentation (MLTS) might lose track of what is behind it, compromising the overall estimation.

- Future work should include the assessment of additional sensors for scan matching to enhance the amount and quality of information available. Additionally, the influence of 3D sensor performance (accuracy, precision, resolution) and their effects on scan matching accuracy should be further explored.

7. Conclusions

In this work, we focused on the localization of autonomous machinery based on the registration of 3D point clouds acquired with a mobile terrestrial laser scanner (MTLS). The point clouds were acquired with a vertical mounting of the sensor, which is common for phenotyping activities; thus, we aimed to maximize the use of information while reducing the equipment associated with localization. We have analyzed four state-of-the-art scan matching approaches: ICP_{pt-pt}, ICP_{pt-pl}, $GICP$, and NDT. Such approaches were used to perform a multi-level registration based on different orchard point cloud information segments. In addition, an analysis of the overlapping between consecutive frames was presented. The results for both the segmentation and overlapping suggest that salient feature structures of the orchard point cloud, with a certain overlap between frames, can increase the registration procedure accuracy, therefore, a better positioning system. The methodology was validated in a Fuji apple orchard, covering different rows within the field. The results show that the root means square error after registration, $RMSE$, was reduced in all the cases, leading to the position error, E_p, to accumulate less error. A significant improvement in localization for all the approaches was appreciated, finding the best response in horizontal segmentation using the $GICP$ and ICP_{pt-pt} approaches. In addition, the translation and rotation errors were analyzed, showing a significant improvement with respect to the original implementations. Even though our approach has shown that it can complement GNSS navigation in the middle-long paths, further studies should be carried out before using this approach as a self-positioning system. For example, the variability of the orchard layout might affect the accuracy of the approach. Additionally, further pre-processing might be needed to remove moving points, since orchards are subjected to be affected by wind. Nevertheless, the fact of dividing the point cloud to improve the localization might have industrial applications, since it could be used for localization of tractors in GNSS-denied environments. The results show herein are encouraging and motivate its usage in a larger context of the localization approach, such as Graph-based SLAM, thus leading the future work will of the authors.

CRediT authorship contribution statement

Javier Guevara: Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jordi Gené-Mola:** Validation, Software, Methodology, Formal analysis, Data curation. **Eduard**

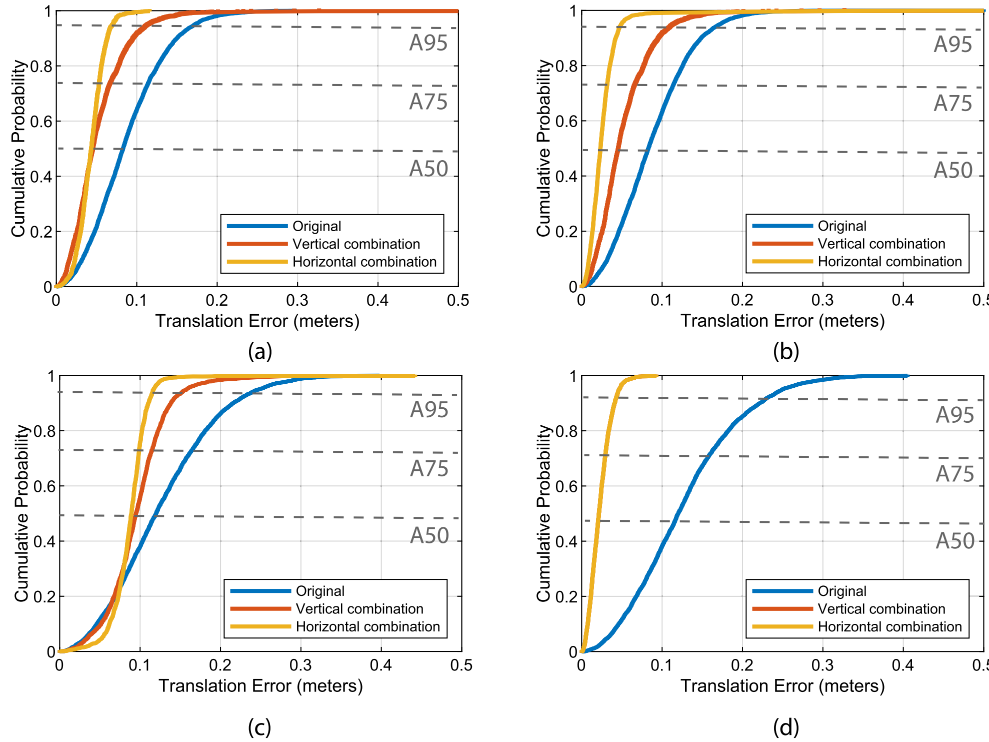

Fig. 18. Cumulative probability of errors for translation. (a) ICP point-to-point, (b) ICP point-to-plane, (c) GICP, and (d) NDT. The quantiles of interest A50, A75, A95 are depicted with a gray line.

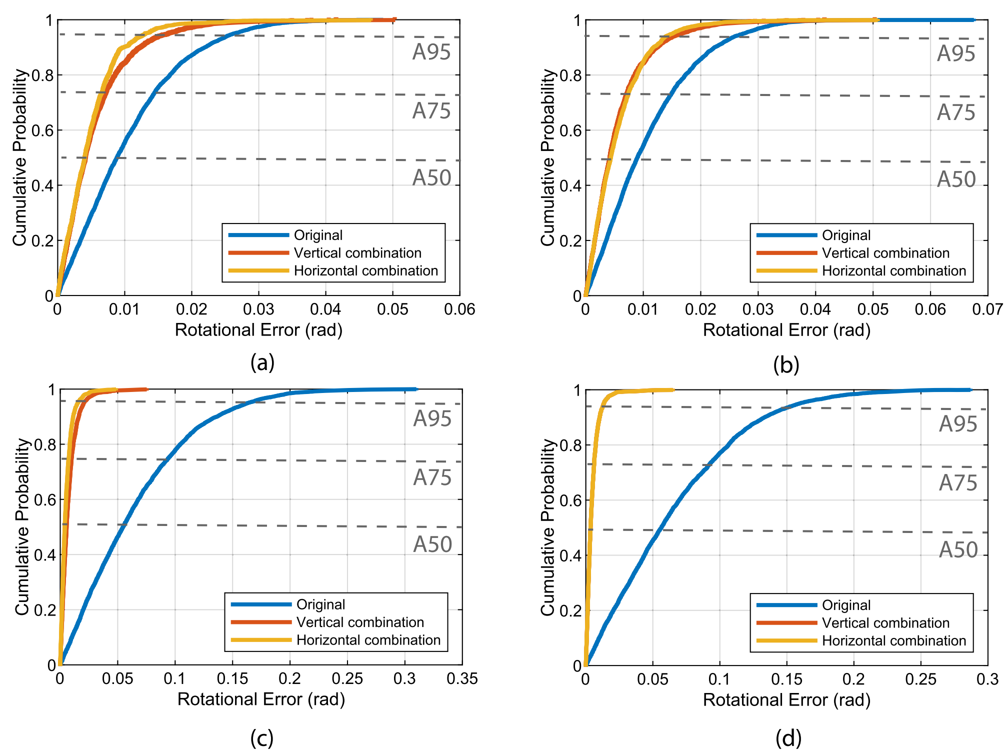

Fig. 19. Cumulative probability of errors for rotation. (a) ICP point-to-point, (b) ICP point-to-plane, (c) GICP, and (d) NDT. The quantiles of interest A50, A75, A95 are depicted with a gray line.

Gregorio: Writing – original draft, Supervision, Resources, Data curation, Conceptualization. **Fernando A. Auat Cheein:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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